

GERALDTON, ON, Canada

WMO: 718340

Lat: 49.783N

Lon: 86.931W

Elev: 348

StdP: 97.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-31.9	-39.6	0.1	-35.0	-36.3	0.1	-31.7	8.9	-10.7	8.3	-11.6	1.0	250	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	28.1	19.3	26.4	18.1	24.9	17.3	20.8	25.9	19.6	24.2	18.5	22.8	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.5	23.4	17.9	13.4	21.9	16.9	12.6	20.7	61.9	25.8	57.6	24.3	54.0	22.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	8.0	7.1	DB	-40.2	31.3	3.6	1.9	-42.8	32.6	-45.0	33.7	-47.0	34.8	-49.7	36.2	(4)
(5)			WB	-40.2	23.0	3.5	1.0	-42.7	23.7	-44.8	24.3	-46.7	24.9	-49.3	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-17.7	-15.6	-8.2	0.2	8.4	14.9	17.6	16.3	11.4	4.0	-4.6	-12.9	(6)	
	DBStd	13.65	8.10	7.54	7.74	5.76	4.94	4.09	3.33	3.44	4.62	4.79	6.66	7.84	(7)	
	HDD10.0	3918	859	716	566	297	91	8	1	1	37	197	437	709	(8)	
	HDD18.3	6305	1117	949	823	545	309	115	54	82	211	445	687	967	(9)	
	CDD10.0	724	0	0	1	2	42	156	236	197	81	11	0	0	(10)	
	CDD18.3	69	0	0	0	0	2	13	31	19	4	0	0	0	(11)	
	CDH23.3	708	0	0	0	1	38	173	290	174	32	1	0	0	(12)	
	CDH26.7	131	0	0	0	0	6	36	58	28	3	0	0	0	(13)	
	WSAvg	3.1	2.9	3.0	3.2	3.3	3.4	3.1	3.1	3.0	3.1	3.3	3.2	2.9	(14)	
	PrecAvg	756	43	31	40	51	74	91	85	82	86	72	57	43	(15)	
(16) Precipitation	PrecMax	1057	121	54	78	113	145	203	178	162	179	130	121	74	(16)	
	PrecMin	577	18	14	22	19	27	28	28	27	13	18	29	23	(17)	
	PrecStd	115	20	11	13	19	30	39	40	37	44	34	21	14	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	5.1	13.4	19.5	27.0	30.1	30.6	29.5	26.6	20.6	12.2	3.6	(19)	
		MCWB	0.6	3.2	8.9	10.4	15.9	19.8	21.1	20.7	18.7	14.2	8.9	2.3	(20)	
	2%	DB	0.1	1.7	8.5	15.4	23.8	27.2	28.2	27.1	23.7	17.0	8.7	1.9	(21)	
		MCWB	-0.5	0.2	4.8	7.7	14.5	17.8	19.7	19.2	17.5	12.9	6.7	0.8	(22)	
	5%	DB	-1.7	-0.7	5.3	12.6	21.1	25.2	26.4	25.3	21.2	14.2	6.1	0.3	(23)	
		MCWB	-2.5	-1.9	2.1	6.0	12.8	16.6	18.4	18.2	16.1	11.3	4.4	-0.4	(24)	
	10%	DB	-4.5	-3.1	2.7	9.7	18.1	23.2	24.8	23.3	19.0	11.8	4.1	-1.3	(25)	
		MCWB	-5.3	-4.2	0.5	4.3	11.2	15.6	17.5	17.2	15.3	9.4	3.0	-2.0	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	2.9	9.7	11.3	18.5	21.6	22.7	22.1	20.2	15.9	9.9	2.7	(27)	
		MCDB	1.3	4.4	12.6	17.6	24.0	28.8	28.7	27.0	24.5	19.1	11.3	3.3	(28)	
	2%	WB	-0.4	0.3	5.1	8.3	15.6	19.4	21.0	20.5	18.5	13.5	6.9	1.0	(29)	
		MCDB	0.2	1.5	7.8	14.4	21.3	25.0	26.1	25.2	21.9	16.3	8.5	1.6	(30)	
	5%	WB	-2.5	-1.8	2.5	6.6	13.8	17.9	19.6	19.3	17.2	11.5	4.7	-0.3	(31)	
		MCDB	-1.7	-0.8	4.6	11.3	19.5	22.2	24.4	23.5	20.0	13.9	5.8	0.3	(32)	
	10%	WB	-5.3	-4.2	0.6	4.8	12.0	16.7	18.5	18.2	15.7	9.4	2.9	-2.0	(33)	
		MCDB	-4.5	-3.1	2.6	9.6	17.1	21.3	22.9	22.2	18.6	11.6	4.1	-1.3	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	11.0	13.1	13.4	12.3	12.5	12.1	11.5	11.1	10.3	7.7	7.5	9.1	(35)	
		MCDBR	9.7	12.0	13.3	17.2	17.8	15.6	13.8	13.8	13.4	12.1	9.4	8.5	(36)	
		MCWBR	9.2	10.5	9.8	10.5	9.5	7.9	7.0	7.3	8.0	8.1	7.4	7.9	(37)	
	5% WB	MCDBR	9.2	10.6	11.4	15.2	14.2	12.4	11.9	11.7	10.8	11.0	8.4	8.5	(38)	
		MCWBR	8.8	9.4	8.7	9.8	8.3	7.1	6.7	6.8	7.3	8.0	6.9	8.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.259	0.296	0.346	0.351	0.366	0.389	0.366	0.347	0.306	0.280	0.272	(40)	
	taud		2.334	2.315	2.305	2.335	2.426	2.411	2.371	2.437	2.490	2.574	2.497	2.271	(41)	
	Ebn at Noon		817	898	914	899	907	893	864	869	853	840	779	712	(42)	
	Edh at Noon		65	85	104	114	109	112	115	103	88	68	57	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.18	2.30	3.65	4.85	5.20	5.71	5.62	4.70	3.33	1.95	1.00	0.82	(44)	
	RadStd		0.14	0.17	0.27	0.39	0.34	0.43	0.40	0.37	0.27	0.28	0.11	0.12	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page